B1 · controlled compliance, fixed readout



Original hard frame · modified TPU · four magnets and plugs · five sensor heads · fixed spine

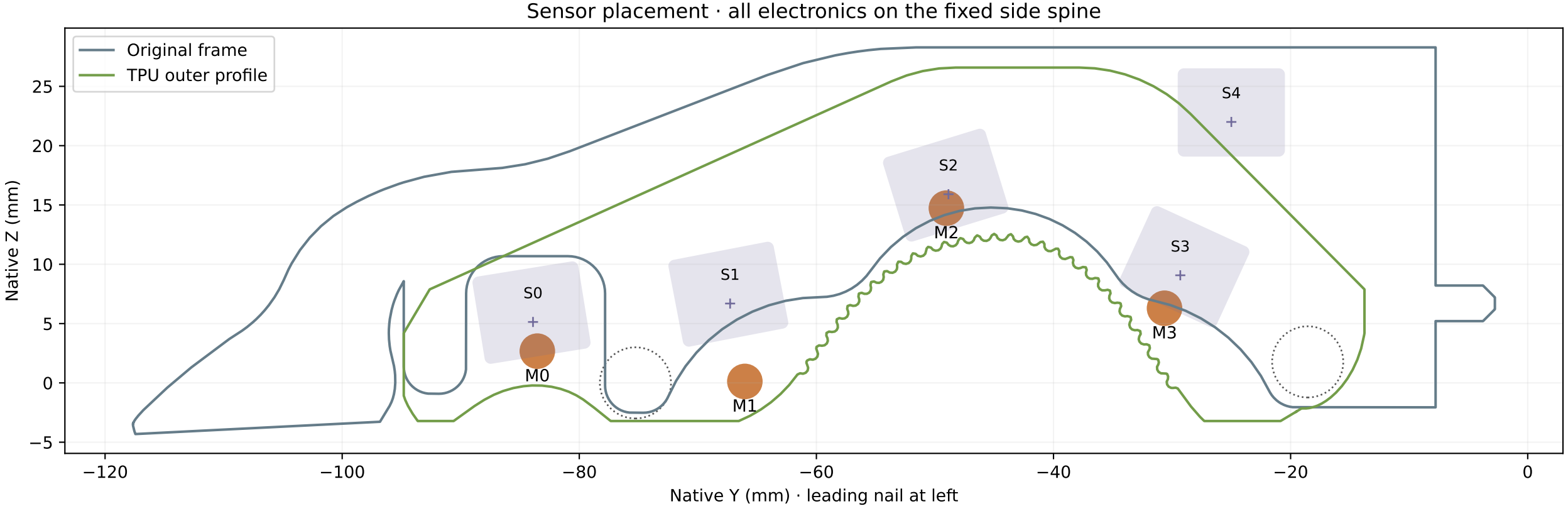
Actual meshes / unbuilt prototype. Wire routing, epoxy fillets and optional nail films are not pictured.

B1 · separate parts



Original hard frame · modified TPU · four magnets and plugs · five sensor heads · fixed spine

Actual meshes / unbuilt prototype. Wire routing, epoxy fillets and optional nail films are not pictured.


Sensor placement · all electronics on the fixed side spine
Original frame
TPU outer profile
S0
S1
S2
S3
S4
M0
M1
M2
M3
Native Z (mm)
Native Y (mm) · leading nail at left
25
20
15
10
5
0
−5
−120
−100
−80
−60
−40
−20
0

B1.0 manufacturing review · millimeters · do not scale this page

Original TPU outer profile and attachment bores retained; hard frame unchanged.
Four side pockets: diameter 3.15, depth 2.15.
Magnets: N48, 3 × 1, axial north toward +native X (sensor side).
Plug: 1.15 long, 3.10 stem, 3.40 bead. Groove 3.55 × 0.30.
Print and gauge a pocket first. Do not force an oversize magnet into TPU.

Pocket centers (X is magnet center, not side opening):
  M0: (-1.650, -83.540, 2.676)
  M1: (-1.650, -66.035, 0.122)
  M2: (-1.650, -49.035, 14.732)
  M3: (-1.650, -30.635, 6.292)

Five identical PCB heads: 8 × 6.4 × 0.8; component face points inward.
PCB component-side plane: X=8.439.
Outer spine: X=10.539. Backing tape: 0.10.
Added global width: 3.65 without heads;
  6.05 with reserved heads; wires excluded.
Reserved M3 head: diameter 6 × height 2.4. Choose actual nonmagnetic hardware.
Carrier bores: 3.4. Measure the original stack to select screw lengths.

No unchanged-stiffness, 20 N capacity, accuracy or fatigue claim.
Use STEP/STL and the PCB fabrication archive; test one build before scaling.